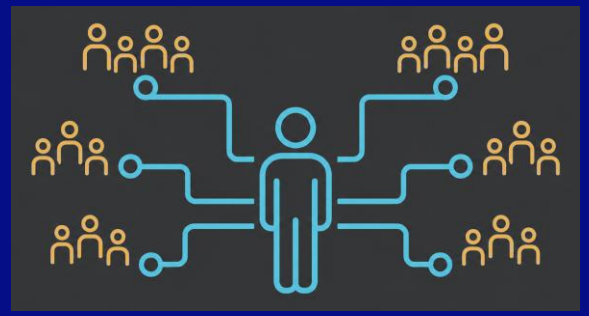


DHUM101
Socially Responsible Computing

University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

Lecture 9
Ethical Frameworks

- Hunter Hatfield
Semester 2, 2026

Today

- You Really Do Care
- Systems of Analysis
- Activity
- What Should You Do?

You Really Do Care

You Really Do Care

This is a required paper. 90% of you must take this paper.

You Really Do Care

This is a required paper. 90% of you must take this paper.

Digital Technologies are affecting you and those you care about

This paper supports you in meeting your own goals.

You Really Do Care

Why are you taking papers in computer science, data science, software engineering?

You Really Do Care

Why are you taking papers in computer science, data science, software engineering?

- It's fun creating things
- You have an idea for a thing you want to exist
- You want to get a job / make money
- You are forced into it

You Really Do Care

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Any others?

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You???

You???

Your Vision For Yourself

I want to create things ~~that hurt people~~

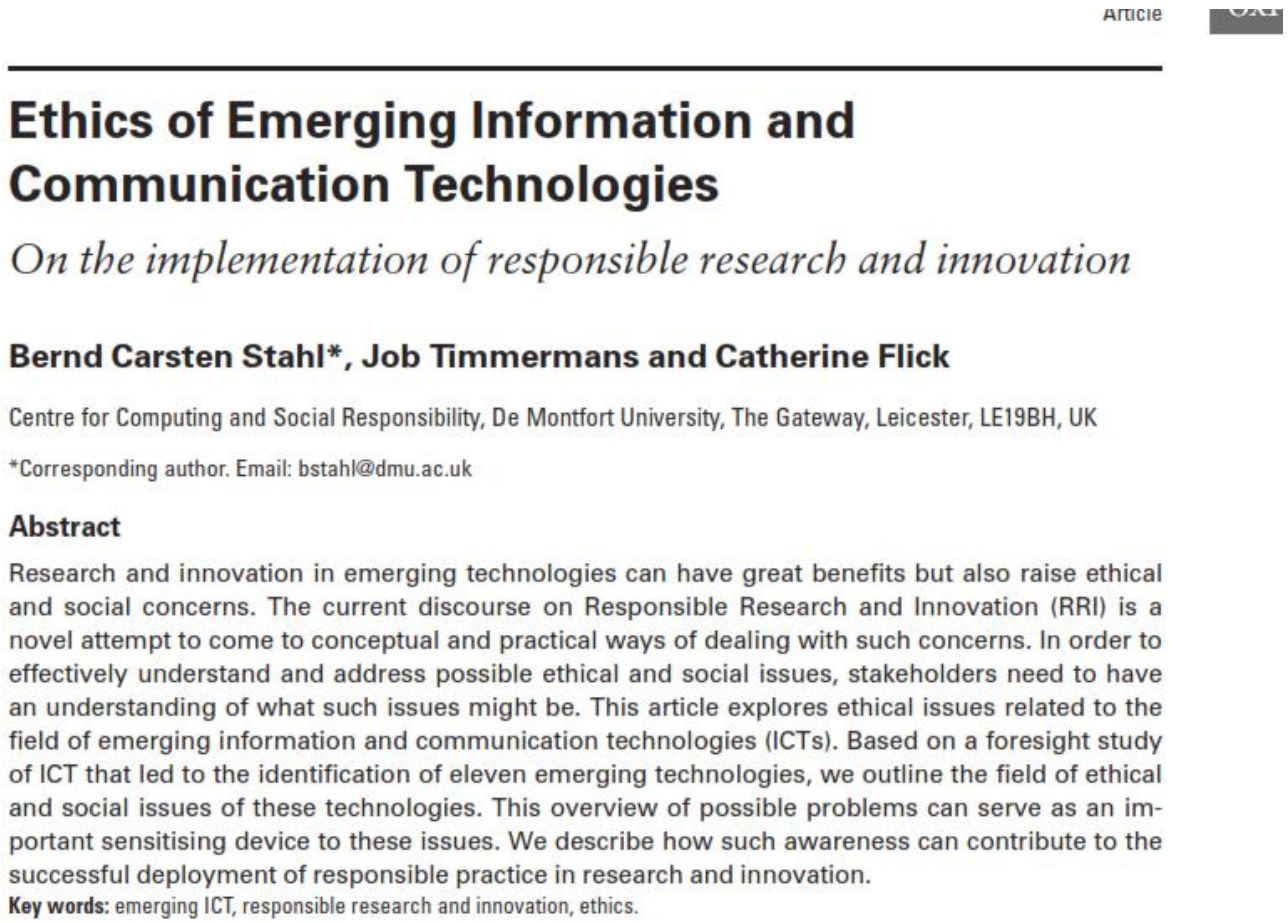
I want to make money ~~by doing things I believe are wrong~~

Takes: Thought, Practice, Learning

Systems of Analysis

A System to Work Through

- Stahl et al's Implementation of responsible research and innovation



Article

Ethics of Emerging Information and Communication Technologies

On the implementation of responsible research and innovation

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Abstract

Research and innovation in emerging technologies can have great benefits but also raise ethical and social concerns. The current discourse on Responsible Research and Innovation (RRI) is a novel attempt to come to conceptual and practical ways of dealing with such concerns. In order to effectively understand and address possible ethical and social issues, stakeholders need to have an understanding of what such issues might be. This article explores ethical issues related to the field of emerging information and communication technologies (ICTs). Based on a foresight study of ICT that led to the identification of eleven emerging technologies, we outline the field of ethical and social issues of these technologies. This overview of possible problems can serve as an important sensitising device to these issues. We describe how such awareness can contribute to the successful deployment of responsible practice in research and innovation.

Key words: emerging ICT, responsible research and innovation, ethics.

Stahl's Responsible Research and Innovation (RRI)

Six Categories.

- Conceptual issues and ethical theories
- Impact on individuals
- Social consequences
- Uncertainty of outcomes
- Perceptions of technology
- Role of humans

Stahl's Responsible Research and Innovation (RRI)

Conceptual issues and ethical theories

Guiding Questions

- Are concepts and terminology regarding the technology already established? How much disagreement exists concerning the scope of the technology?
- Does current research concerning the technology cross academic discipline boundaries? If so, are there problematic definitions of terms and concepts across these boundaries?
- Do ethical questions concerning the technology imply a particular ethical position?

Stahl's Responsible Research and Innovation (RRI)

Conceptual issues and ethical theories

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Get an understanding of the key ideas from both ethics and related research

Don't start from scratch

Stahl's Responsible Research and Innovation (RRI)

Impact on Individuals

- Privacy
- Autonomy
- Treatment of humans – actual impact on people
- Identity
- Security

Stahl's Responsible Research and Innovation (RRI)

Impact on Individuals: Autonomy

- In what ways does the technology improve independence/autonomy?
- To what extent does the technology monitor or control people's behaviour, attitude, emotions, thoughts, moods, and actions?
- Does the technology make decisions? What are these decisions based on, and do they take into account ethical issues

Stahl's Responsible Research and Innovation (RRI)

Impact on Individuals: Treatment of Humans

- In what ways could the technology impact on the daily life of people?
- Could vulnerable people be particularly affected by this technology?
- Does the technology seek informed consent where necessary?
- Could events that happen within the virtual world of the technology negatively impact on the real world?

Stahl's Responsible Research and Innovation (RRI)

Impact on Individuals: Identity

- Does the technology change human capabilities, e.g. their ability to perform certain tasks?
- How will the technology affect the way in which users see themselves or one another?

Stahl's Responsible Research and Innovation (RRI)

Social Consequences

- Digital divides – what peoples have access to
- Collective human identity and the good life – effects on practicing cultures
- Ownership, data control, and intellectual property
- Responsibility – who takes responsibility for actions and consequences?
- Surveillance
- Cultural differences

Stahl's Responsible Research and Innovation (RRI)

Social Consequences: Collective human identity

- Does the technology replace established human activities or work?
- Which view of culture or human society is the technology likely to promote?

Stahl's Responsible Research and Innovation (RRI)

Social Consequences: Responsibility

- Which existing and legacy system does the technology rely on?
- Who is responsible for testing of the system?
- Which consequences could a malfunction or misuse of the technology have?

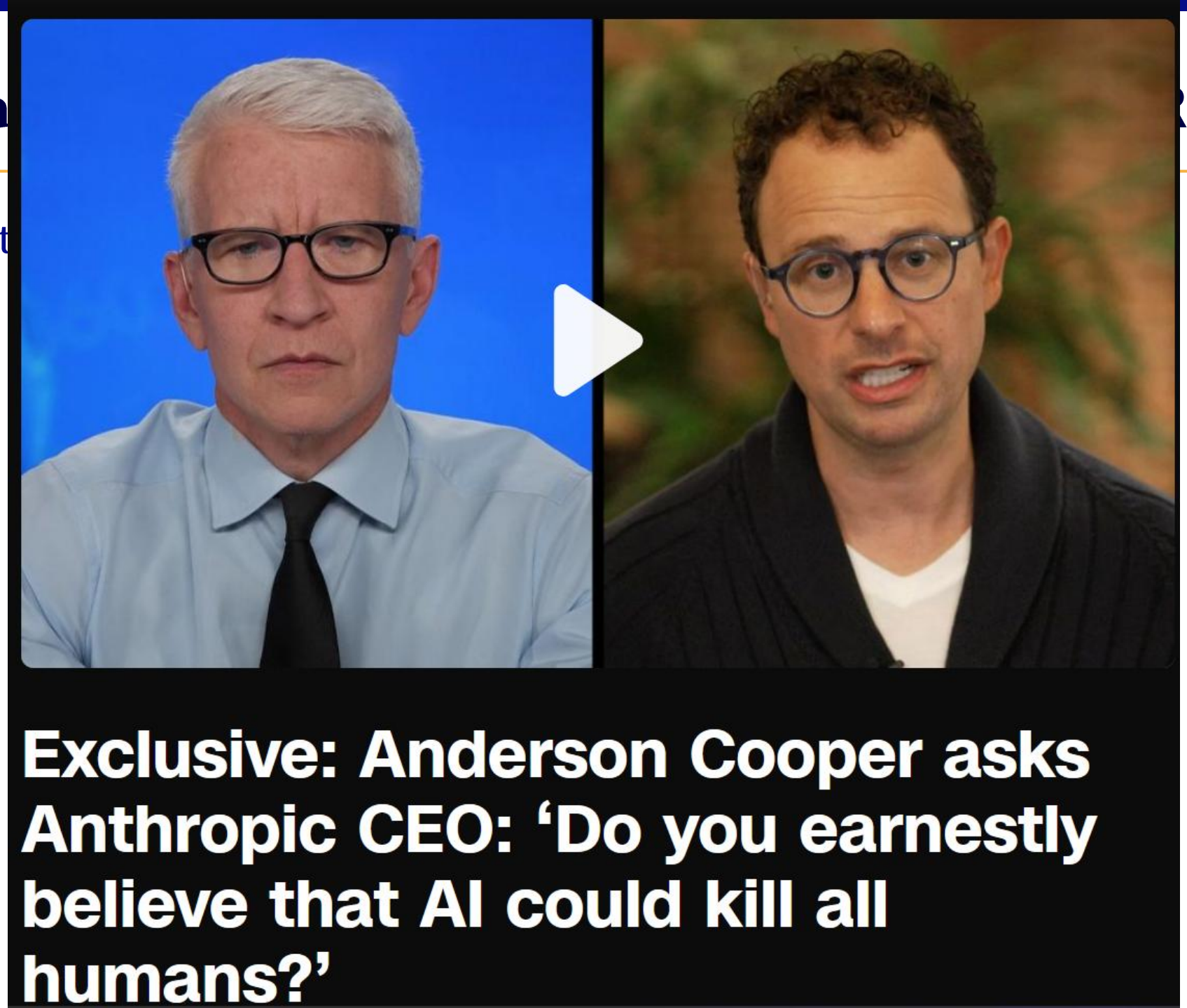
Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes



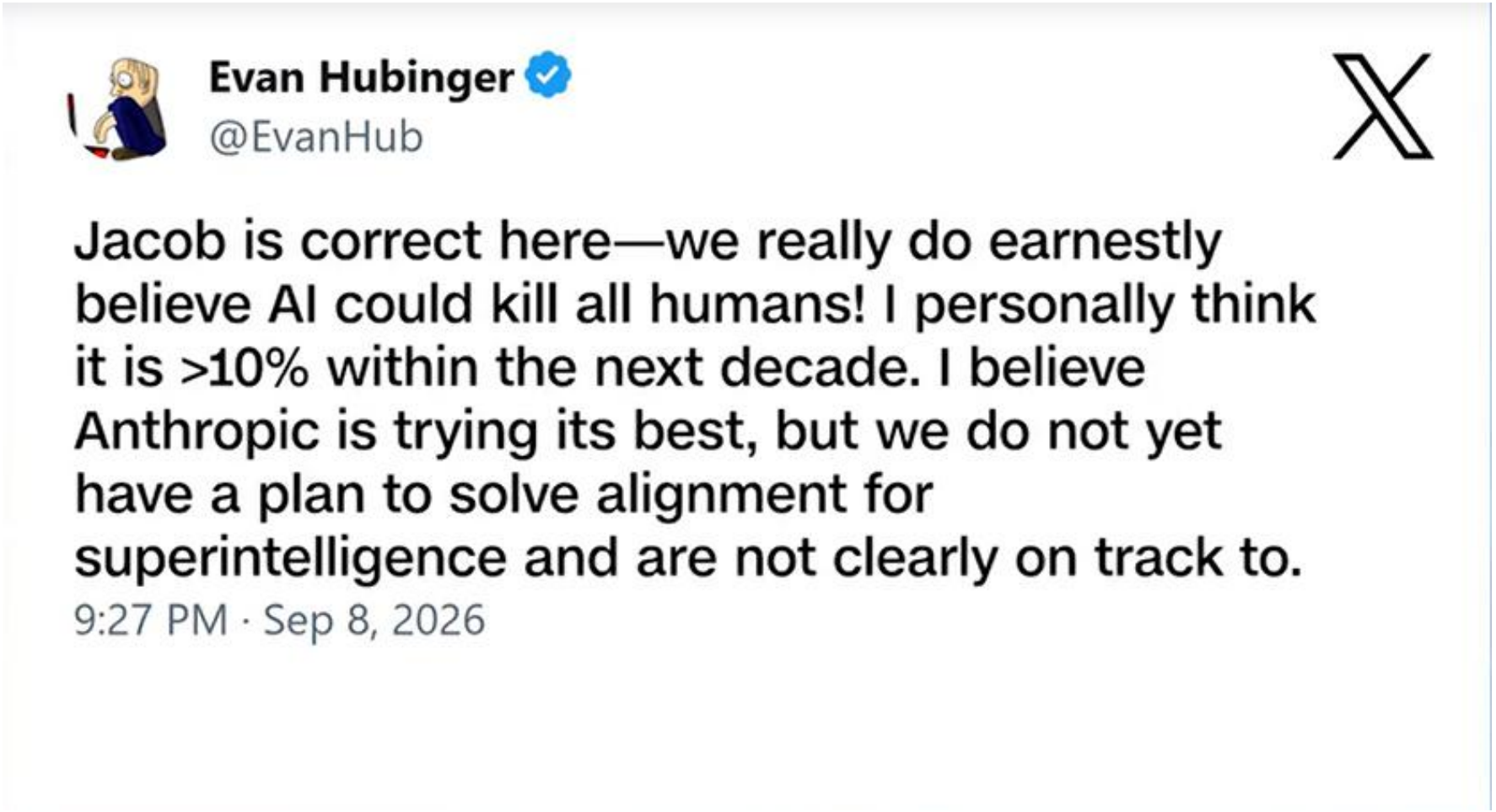
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Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes



Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes Long-term future positive outcomes justify short-term negative outcomes

Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes Long-term future positive outcomes justify short-term negative outcomes



Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes Long-term future positive outcomes justify short-term negative outcomes

Argument:

- People have been limited by information processing
- General AI will make information processing practically unlimited
- In the future, people will do amazing good things with information processing
- Conclusion: Any limitations to this future is unethical

Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes Long-term (very uncertain) future positive outcomes justify short-term (very certain) negative outcomes

Also assumes full-blown consequentialism

Stahl's Responsible Research and Innovation (RRI)

Uncertainty of Outcomes (Guiding Questions)

- What are the possible uses of the technology beyond the ones primarily envisaged?
- Are there foreseeable side effects or unintended consequences of the technology?

Stahl's Responsible Research and Innovation (RRI)

Perceptions of Technology

- Will the technology appear autonomous to users?
- Will the technology be anthropomorphic, that is, look or act in ways that we normally expect humans to look or behave?
- Which human activities will be replaced by the technology?

Why is this important?

Stahl's Responsible Research and Innovation (RRI)

Perceptions of Technology

- Will the technology appear autonomous to users?
- Will the technology be anthropomorphic, that is, look or act in ways that we normally expect humans to look or behave?
- Which human activities will be replaced by the technology?

Why is this important?

- It is challenging to separate human-like and human (personhood)
- Actual persons have moral worth

Stahl's Responsible Research and Innovation (RRI)

Role of Humans

- Which novel capabilities will the technology provide users with?
- Will the technology be closely linked to the user (e.g. be wearable) or implanted?
- Is the technology likely to replace established human activities or work?

Why is this important?

Stahl's Responsible Research and Innovation (RRI)

Role of Humans

- Which novel capabilities will the technology provide users with?
- Will the technology be closely linked to the user (e.g. be wearable) or implanted?
- Is the technology likely to replace established human activities or work?

Why is this important?

- Changes how people relate to each other and live our lives

Stahl's Responsible Research and Innovation (RRI)

Six Categories.

- Conceptual issues and ethical theories
- Impact on individuals
- Social consequences
- Uncertainty of outcomes
- Perceptions of technology
- Role of humans

Goals:
Comprehensive:
39 Guiding Questions

Doesn't Answer:
Only guides

Activity

Activity

Technology: A Large Language Model that can code and release Large Language Models

- Are concepts and terminology regarding the technology already established? How much disagreement exists concerning the scope of the technology?
- Who is responsible for testing of the system?
- How will the technology affect the way in which users see themselves or one another?
- Are there foreseeable side effects or unintended consequences of the technology?

Let's Apply It!

What should you do?

What Should You Do?

Your goal is to take actions that match your goals

Don’t start from scratch

- Explore systems that have already been created (such as RRI)
- Find conversations about digital development and engage with them

Don’t let the perfect be the enemy of the good

- “I can’t do everything right, so I’ll just ignore it.”
- One step forward is better than no steps forward

Look for people and systems that ask good questions

- You are looking for guides, not dictators
- But listen to experts

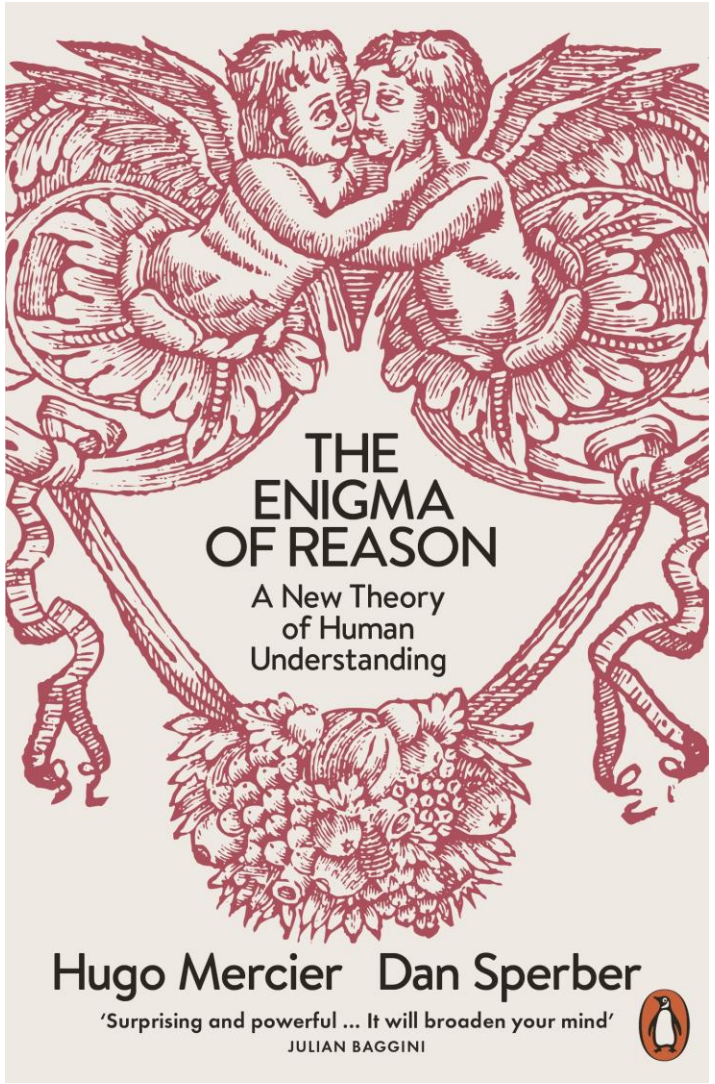
What Should You Do?

- Reflection
- Cooperation
- Process

Cooperation

Humans cooperate

- We can do more together than apart
- Fundamental to human activity
- Single people don't sail to the stars
- Groups of people do



Cooperation

How do we cooperate?

- We divide up tasks
- We have expectations for one another
- These tasks and expectations allow us to accomplish shared goals.
- Therefore
  - We need to communicate to others what we believe and why
  - We need to justify our actions to others
- But
  - We do not want to fall victim to others cheating, taking, not contributing
  - So we scrutinise them carefully

Cooperation

You will make your own errors and indulge your own processes unless you cooperate

- Diverse team
- Diverse experts
- Diverse research

mplish shared

and why

ng, not

Process

Stages of Development.

- Planning
- Building
- Deploying
- Monitoring

Consider both benefits and harms at all stages.

Next Time

Conclusions

Technology for good!